

Universiteit van Amsterdam

Identifying workplace accidents in news articles

Erik Meijer
Rory Guliker
Taiki Papandreou
Thomas Meeusen



Universiteit van Amsterdam
The Netherlands
June 2022

Contents

1 Introduction **2**

2 Problem Description **2**
2.1 The Client 2
2.2 The Problem 2
2.3 Problem Vision 3

3 Implementation **3**
3.1 Dataset 3
3.1.1 Data 3
3.1.2 Collocation 4
3.1.3 Labeling 4
3.1.4 Clustering 5
3.2 Models 6

4 Results **10**
4.1 Lbl2vec 10
4.2 Logistic Regression 10
4.3 Label Propagation 10
4.4 BERT 10

5 Discussion **12**
5.1 Conclusion 12
5.2 Future Work 12

6 Appendix **13**

References **13**

1 Introduction

They say that there is a fine line between success and disaster, and that is certainly not inaccurate. Accidents happen daily everywhere in the world, potentially causing damages and injuries wherever they occur. As such, it has always been a global effort to analyse how these occur, and how to stop them. The dawn of artificial intelligence and machine learning as a whole has provided us with new opportunities to do so. By using techniques in these fields, we can analyse patterns in existing data, such as articles in the news, much faster, leading to potential discoveries of new issues, and new solutions.

This paper describes an experimental project assigned by the company TNO, in order to analyse existing news articles to detect cases of specific types of accidents, so that this information may later be used by the stakeholders themselves to detect hidden trends and patterns. Subjects of discussion will be the specific details of the proposed project, initial solutions, the used data, implemented models, and the final results of said models.

2 Problem Description

2.1 The Client

This project was proposed by Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek (TNO). TNO is an independent research organisation in the Netherlands with offices worldwide that enables businesses and governments to apply scientific knowledge in practice. It was founded in 1932 and it has 3,652 professionals working on various projects that concern fields such as Artificial Intelligence, security, energy transition and infrastructure.

2.2 The Problem

As part of a larger project being developed by TNO, the stakeholder has assigned the development team to the following component in their program: The goal was to develop a methodology that was intended to use Natural Language Processing (NLP) methods to analyse datasets of potentially hundreds of thousands of news articles. This would then be used as training data for a classifier to predict a distinguishment between articles regarding accidents as a result of occupational hazards, and articles not regarding these. If possible, it was also requested that a potential deeper layer of classification would be utilized in order to differentiate between specific types of accidents on a professional capacity (in this paper commonly referred to as 'workplace accidents').

The base principle of this assignment would be to experiment with various different models and methods, and report our findings on which were most successful back to the stakeholder. Upon the completion of the assignment, the stakeholder would then proceed to apply our methodology on different datasets

of their own and potentially integrate this into their wider project. That being said, considering the rather specific nature of the classification labels required for this project, proper labelled data on the subject would be very difficult to come by, if it existed at all. Therefore, a large portion of the task included finding ways to properly (and as accurately as possible) label datasets, so the stakeholder could utilize this method on other datasets than the ones used here. If this effort was not succesful, alternatives in unsupervised learning would have needed to be considered.

2.3 Problem Vision

Due to the experimental nature of the project, combined with the very specific classification requirements, it was decided early on to divide the project up into three stages. Stage 1 would be to classify general accidents, stage 2 would go on to classify workplace-accidents specifically, and finally stage 3 would implement the final phase of differentiating between different kinds of these workplace accidents. In order to achieve these stages, manual labeling would be required to a certain degree. As such, by utilizing a list of keywords, a chunk of data entries would be labelled as negative for accidents, with the remainder of the data subset being labelled personally. Once a labelled set (of approximately 10000 entries) was usable, the plan was to train supervised or semi-supervised models on the set for the first stage of precision to isolate general accidents. The second stage could largely be done in a similar fashion, now using the most successful model from stage 1 to isolate a data subset to use in labelling, then trialling a series of models on these entries. These results would then be further divided into different categories, presumably through the use of unsupervised clustering algorithms for stage 3.

For the project, the stakeholder had specifically requested the trialling of multiple different algorithms. As such, rather than using one specific model for each stage, several would be implemented and tested for their accuracy for each of the stages.

3 Implementation

3.1 Dataset

3.1.1 Data

The used data is the CNN dailymail dataset found on Hugging Face[1]. This dataset contains more than 300.000 news articles about different kinds of mostly negative events, such as accidents, disasters, attacks, and even celebrity gossip. Unfortunately, it was not initially labelled and many articles did not concern accidents (as described by the stakeholder's requirements) at all. As such, before

[1] `https://huggingface.co/datasets/cnn_dailymail`

we could reach workplace accidents, we first had to isolate accidents in the broader term, and started binarily classifying the dataset into non-accident and accident-classes, technically referred to as class 0 and 1 respectively.

3.1.2 Collocation

In order to find more words that could be used as keywords to identify potentially relevant articles, we developed a tool to quickly find which word frequently co-occur in the dataset. The dataset was preprocessed by encoding each article as an array of word IDs, so that the entire dataset can be searched in a few seconds. The function counts all words that occur around the keyword in a context word of configurable size. The score of each word is then multiplied by the inverse document frequency to remove uninformative words such as 'the'.

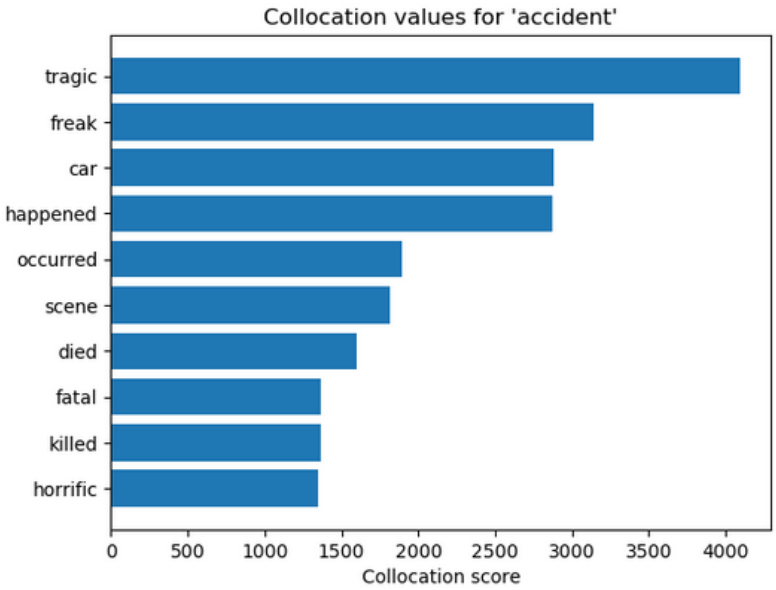


Figure 1: Most frequently co-occuring words in a window of 5 words, corrected by the inverse document frequency.

3.1.3 Labeling

Considering the size of the dataset, with over 300.000 articles, they could not be all manually labelled. Instead, subsets of the data were taken using both positive and negative keywords, with those not adhering to this labelled as 0. The remaining subset were considered candidates for accidents, and were labelled manually, with the criteria of being a spontaneous, non-intended accident placing specific people in harm's way. Incidents such as shootings for example, were thus classified as non-accidents, as they were intentional. Natural disasters were

an important distinction to make, as they could sometimes be classified as either class. As a rule of thumb, articles covering the broad natural disaster itself were classed as 0, while articles regarding the misfortunes of specific scenarios during- or as a result of the disaster were classed as 1. An interesting discovery during the manual labelling, was that many of the articles pointed at the same event, for example, multiple articles were found about the same earthquake or about Michael Jackson's cardiac arrest. This pointed at the conclusion that most of the articles were divided in only a small set of categories for events, which made the quality of the dataset less useful, unfortunately.

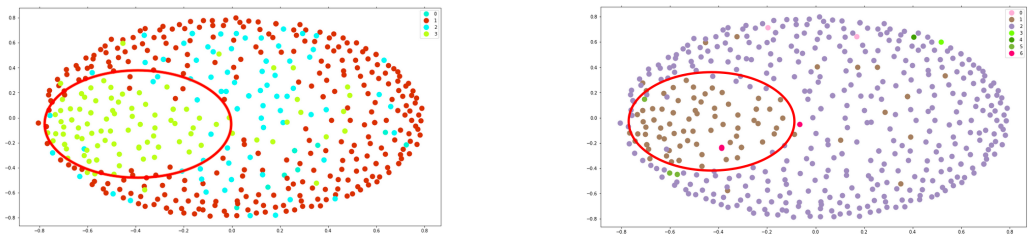
When a satisfactory model was trained for Stage 1, this model was used to perform a similar strategy for stage 2, the identification of workplace accidents specifically. First, using the model to filter out non-accidents, a list of candidates, consisting of accident-class articles, was assembled. Then, once again, manual labelling proceeded on this subset. This was again done in a binary fashion, with the new 'class 0' being non-workplace-accidents (including the very rare non-accidents that slipped through the algorithm), and 'class 1' being workplace accidents. The criteria for these were quite simple: An un-intended incident placing real people in harm's way, taking place while these people are physically at work performing their duties.

After the labelling in both stages, a concerning albeit predictable (considering the varied nature of articles) issue was uncovered. There was a massive class imbalance between class 0 and 1 in both stages 1 and 2. Both times, only about 10 percent of all manually labelled articles would be classified as positive. This would later cause issues for most tested algorithms, with artificial class balancing being mostly unsuccessful, with one major exception.

3.1.4 Clustering

One of the subtasks of TNO was to find words that describe various categories of news articles.For instance, it stood out during the manual classification work how many articles regarding accidents connected to airline crashes, typhoons, and the earthquake in Haiti. TNO asked us to look for keywords that may be utilized to group these various subjects together.
Data were represented as a tf-idf matrix in order to accomplish this, and articles were clustered using the k-means algorithm. Because nouns exhibit the most characteristics of articles, articles were divided into nouns, which greatly increased the efficiency of the procedure. After that, K-means algorithm was employed after modeling all of the articles as a tf-idf matrix. The K-means algorithm is a straightforward clustering algorithm that gives labels to nearby data points and selects k cluster beginning points at random.
Decomposed vectors were used to illustrate multidimensional vectors (i.e., rows of the tf-idf matrix) on a two-dimensional plot. Numerous data points were unable to cluster in a meaningful way, as shown in figure 2. The articles relating to aviation crashes were successfully grouped, indicated by red circles in figure

2a and 2b. However, because of the poor result of grouping, we decided against looking any further.



(a) k = 4 (b) k = 7

Figure 2: Our data in k clusters

3.2 Models

Logistic Regression Our first attempt at classifying news articles was using a logistic regression model, which can categorise articles into two classes: articles that are about an accident, and those that are not. Since logistic regression operates on numeric data, we used a simple method to represent a document as a number. A score is calculated for each document using a weighed sum over all words. The weights associated with each word is based on the average tf-idf of that word in both classes. Because this is essentially a bag-of-words model, as it ignores the order of the words, the model did not have great results. Figure 3 shows how the scores assigned to positive and negative documents overlaps, making it impossible for logistic regression to classify the articles correctly 100% of the time.

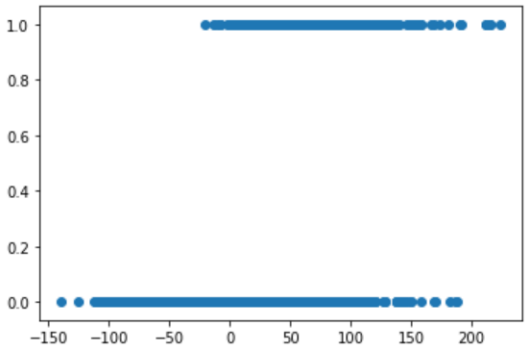


Figure 3: Overlapping scores of positive and negative labels. The horizontal axis is the score assigned to the document, the vertical axis is its true label.

lbl2vec Lbl2vec is a Python Package singly developed by Tim Schopf, a Ph.D candidate at the University of München, Germany. This is an unsupervised text classifier. It clusters documents to a predefined label. This label is defined by the keywords tagged to it. All words and documents in the training set form a vector and the idea is that semantically similar keywords define a cluster. As the figure 4 suggests. (Schopf., Braun., & Matthes., 2021)

Lbl2vec has the advantage that outliers are detected and not used for the

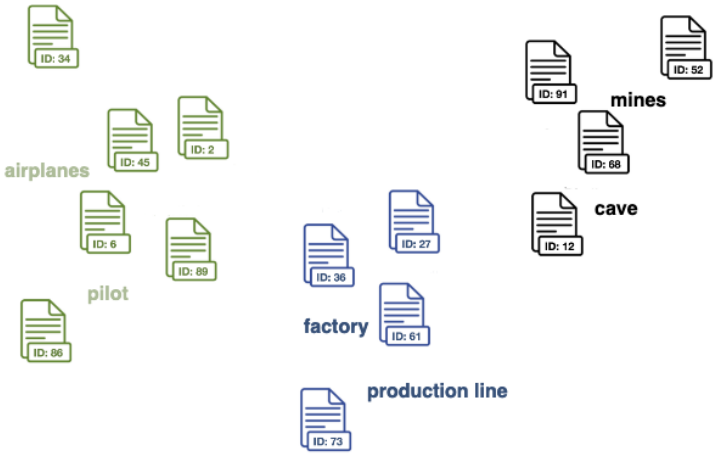


Figure 4: Lbl2vec clusters the documents in different categories, for instance: football, basketball and baseball

calculating of a cluster. After the outliers are removed, centroids in the vector space are calculated. The distance to each datapoint is calculated and assigned to the the label corresponding with the nearest centroid. (Schopf. et al., 2021)

After preprocessing and partly manually labeling the dataset, it was divided in a training set and a test set, with a ratio of 80%-20%. Initially, two labels were used, positive and negative, attached with the most common keywords. However, due to those keywords were not that much, it resulted in a rather low f1 score of 40%. Working with Lbl2vec was a bit of a challenge, since it was developed by one person, it did not bring much documentation with it. Another downside is that it take a long amount of time to train a model.

Label Propagation One of the obstacles of this project was the unlabeled data. Due to them being unlabeled, supervised learning methods were much more difficult to implement. Therefore, a semi-supervised learning method was investigated. The semi-supervised learning method concerns with both labeled and unlabeled data. In a world where good quality data are scarce semi-supervised learning methods still enable us to perform a supervised learning classification to some degree. Label Propagation is one of the semi-supervised machine learning algorithms that assigns labels to previously unlabeled data points. It is based on graph theory and the idea that location of data points indicate similarity between the points. It was developed by Xiaojin Zhu and Zoubin Ghahramani from Carnegie Mellon University in 2002 (Zhur & Ghahramanih, 2002).

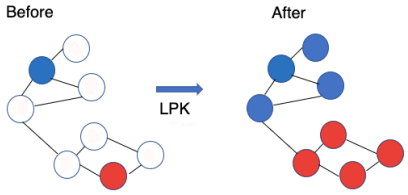


Figure 5: Label Propagation visualized

As the figure 5 suggests data are represented in points (or in vectors). Some data points are colored indicating labels and blank points have no assigned labels. Using clustering algorithm such as LPK-means algorithm the model tries to predict unlabeled data points according to their neighboring labeled data points. The right side of figure 5 shows the result of the clustering algorithm. This way the model manages to predict label of data using both labeled and unlabeled data.
Our data were represented as tf-idf matrix where the rows indicate the documents and the columns indicate the unique nouns that are used in the data. Simply put, each document was numerically represented as a vector of word frequency. Documents which were manually labeled made 50% of the input data and unlabeled data made the other 50%.
After the initial representation of our documents, LPK-means algorithm was applied to guesstimate labels of unlabeled data points.

BERT BERT stands for Bidirectional Encoder Representation from Transformers. BERT was created and published in 2018 by Jacob Devlin and his colleagues from Google (Devlin, Chang, Lee, & Toutanova, 2018). It employs a series of transformer encoders, which learn context of documents. Transformer is a deep learning model that revolutionized the NLP scene. It made it possible for computers to comprehend natural language more effectively. BERT uses bidirectional training of transformer encoders using Auto-encoder language

modelling. Training consists of two parts; pretraining and finetuning. During pretraining BERT tries to learn context by performing two unsupervised learning tasks. These are masked language modelling and next sentence prediction tasks. During finetuning BERT tries to learn how it should use language for specific task. During this step BERT learns to expertise in language that is used in accident related articles.
Figure 6 illustrates our classifier model that used BERT algorithm. The model was developed by TensorFlow. A document is first preprocessed in the required format, and then it is processed by BERT encoders in accordance with that format. This gave each document the context that BERT believes it to be about. The dropout layer is then used to remove extraneous parameters from the BERT output. After that, document was classified using its context defined by BERT. The output of the model is a possibility of the input document being work accident related or not.

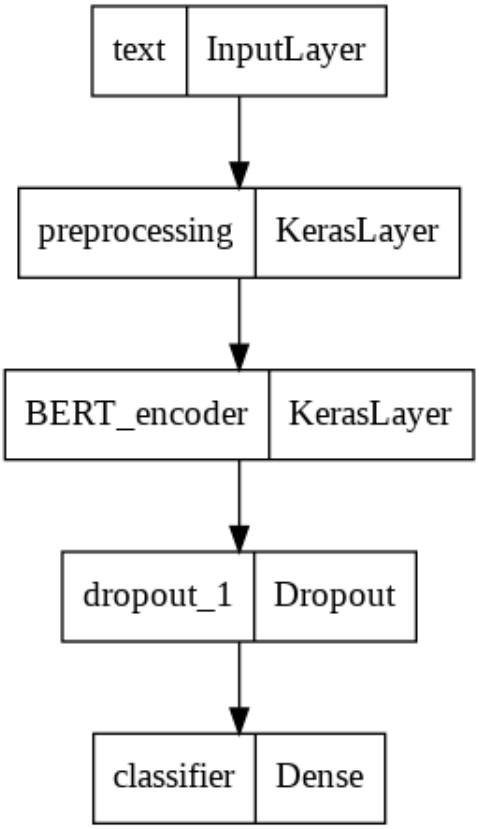


Figure 6: Classifier model using BERT made by TensorFlow

4 Results

4.1 Lbl2vec

The results from the various models, save for one, were somewhat underwhelming. Lbl2vec, as expected from the over-reliance on clearly definitive keywords and the huge disparity in the data, had the low average accuracy of 37 percent for stage 1. Furthermore, the accuracy in specifically identifying general accidents was only 7 percent. These findings practically eliminated the use of this model as a viable method to achieve our objective.

4.2 Logistic Regression

Logistic regression had significantly better results, with the average accuracy turning out to be 79 percent. That being said, the f1 score of classifying as accidents specifically was merely 0.52, while that of non-accidents was 0.86. This was likely yet again due to the large disparity in data between the two classes, as well as the fact that scores overlapped significantly for the two classes considering most of the words used in them would always be fairly similar. Unfortunately, while significantly better than the prior attempt, this score was still considered to be far too unreliable for proper usage. After all, a 50/50 chance of being correct in a binary classification problem is nothing more than simply guessing.

4.3 Label Propagation

The semi-supervised label propagation method fared even better. Its average scores were very impressive, with an average accuracy of approximately 97 percent. That being said, these scores were once more deceptive. The f1-score for the general accident class was only 0.68, in comparison to the non-accident class' 0.99, once more as a result of the small amount of class-1 labels, something even more prevalent in a semi-supervised model. That being said, the 0.68 score was certainly a respectable improvement over the prior models. As such, label propagation was kept as a viable candidate, and remains the second-most accurate model aside from BERT.

4.4 BERT

The final model, BERT, was tested using various different types of sub-models intrinsic to the base design of BERT. All of these, practically from the first tests, were more accurate than anything we had attempted before, with f1-scores for the general accident class (class 1) being roughly 69 percent. The seemingly most accurate version of the model was tested using various different artificial balances in the amount of data per class. The initial stage-1 version, with class

Model	Precision	Recall	f1-score	Accuracy	Weighted Accuracy
Lbl2Vec	0.99(0) 0.07(1)	0.33(0) 0.96(1)	0.49(0) 0.13(1)	0.36	0.47
Logistic Regression	0.81(0) 0.69(1)	0.93(0) 0.42(1)	0.86(0) 0.52(1)	0.79	0.77
Label Propagation	0.98(0) 0.98(1)	1(0) 0.52(1)	0.99(0) 0.68(1)	0.98	0.97
BERT (75/25)	0.82(0) 0.77(1)	0.94(0) 0.49(1)	0.87(0) 0.60(1)	0.81	0.79
BERT (25/75)	0.68(0) 0.82(1)	0.98(0) 0.17(1)	0.80(0) 0.28(1)	0.69	0.61
BERT (50/50)	0.74(0) 0.77(1)	0.77(0) 0.74(1)	0.75(0) 0.76(1)	0.76	0.76
BERT (50/50 L8A12)	0.85(0) 0.90(1)	0.87(0) 0.88(1)	0.86(0) 0.89(1)	0.88	0.88
BERT Stage 2	0.99(0) 0.95(1)	1(0) 0.83(1)	0.99(0) 0.89(1)	0.99	0.99

Table 1: The sklearn-provided classification reports for all individual models. Here, (0) and (1) refers to the scores of the individual classes within a model.

balancing set to 75/25 (for class 0 and 1 respectively) was relatively inaccurate with an average score of 0.60 for accidents specifically, as was now expected from the class imbalance in the favor of non-accidents. When this was switched around however, with a balance of 25/75 in favor of the accident class, accuracy dropped significantly to 0.28 for this class. This was likely due to significant overfitting on the few (replicated) dataset entries that were actually labeled as such. When balancing the classes to 50/50, accuracy practically surged using the L8 A12 submodel. Both classes had now achieved an accuracy of approximately 90 percent, with the class 0 f1-score being 0.86, and the class 1 score actually being higher at an exceptional 0.89. It was decided that this was the model that would be used further, and taken to stage 2. As these scores essentially made the previous models obsolete, their stage 2 experimentation was discontinued, with their further usage in this project not being recommended.

Following manual labelling of more entries for usage in our next phase, the implementation of BERT for stage 2 was handled swiftly, and results proved once again rather conclusive. Without any further manipulation of the data, other than a 80-20 split into training and test data, the prior BERT model proved very reliable with a very high average accuracy of 99 percent. The f1-scores of the classes were 0.89 for the workplace-accident class, and the much higher 0.99 for the non-workplace accident class. This difference was expected, as now the non-accident class once again much outnumbered their counterparts. With a lack of competing alternatives, this implementation was deemed the most superior by far, with its performance being considered satisfactory. It is highly recommended that the stakeholder proceeds with the usage of this particular model.

5 Discussion

5.1 Conclusion

It has become very clear that out of all the trialled models, the BERT algorithm is most suited for the task if provided with proper data. To this end, it is recommended that the stakeholder uses data with as little class imbalance as possible. It should be worth noting that artificial class balancing to a 50/50 ratio seems to be effective for BERT provided that the minority class comprises at least 10 percent of the total training data. The other models also run into significant problems due to class imbalances, however, it seems that these ultimately are unable to compensate for this by artificially balancing the data. This is likely caused due to significant overfitting on the small amount of class-1 data if this is artificially replicated, and BERT seems to be the only trialled model capable of properly handling this. Due to its over-reliance on predetermined keywords, Lbl2vec is not recommended, and logistic regression failed to yield convincing results due to very overlapping scores (as a result of plentiful recurring words).

5.2 Future Work

Due to time constraints, the optional third stage has not yet been completed. That being said, from prior experiences, advice for a future completion of this phase by the stakeholder can be drawn up. Unsupervised clustering algorithms might have proven to be inaccurate thus far, but that does not necessarily mean they are of no further use. Considering stage 3, unlike the prior phases, has more than two possible classes, an extra step could be required in case these possible labels are not immediately clear. A clustering algorithm could be used to not classify, but *identify* groups of specific types of workplace accidents that might not immediately have been considered as a category. Once the classes have been identified, a similar process to stage 1 and 2 could be repeated, taking a subset of workplace-accident articles (as identified by the stage 2 model), and manually labelling these to the found classes. It is recommended that no less than 2000 articles are used at the bare minimum. It is also recommended to the stakeholder to also use more labelled data for stages 1 and 2 than the amount used in this specific project due to limitations in time and resources, to avoid potential overfitting.

Another future improvement could be made by post-training the BERT algorithm on the fully unlabelled data, to allow the model to train itself on the style of text it will be made to process. This is usually already done in BERT models for different kinds of english texts, but could also be performed manually for a more precise skill in these types of (dutch) articles.

6 Appendix

A more detailed documentation of the implemented models, and the source code for our work, can be found in this repository on Google's cloud storage platform 'Google Drive':
`https://drive.google.com/file/d/1FSSYwvVDaB9AK6zmEdv_MS038qpjjDPn/view?usp=sharing`

References

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

Schopf., T., Braun., D., & Matthes., F. (2021). Lbl2vec: An embedding-based approach for unsupervised document retrieval on predefined topics. In *Proceedings of the 17th international conference on web information systems and technologies - webist,* (p. 124-132). SciTePress. doi: 10.5220/0010710300003058

Zhur, X., & Ghahramanih, Z. (2002). Learning from labeled and unlabeled data with label propagation.